AIR RESIDENCE

Visual: project identity graphic generated from the supplied project list. Official project images are available through the source links below.

Why is this project a best practice?

Challenge

The project had to provide high thermal comfort through cold winters and warm summers while sharply reducing operational energy. The main design challenge was to coordinate a high-performance envelope, efficient services, renewable energy and healthy materials without compromising the architectural concept, construction quality or everyday usability.

Solution

The response combines mineral-wool insulation and underfloor heating designed for comfort and reduced demand; photovoltaic panels intended to offset electricity used in common areas; and secure bicycle parking and a compact four-storey urban format. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The core package - demand reduction first, airtight construction, efficient low-temperature systems, ventilation and on-site renewables - is transferable to other Romanian homes. Replication requires climate-specific energy modelling, careful thermal-bridge and airtightness detailing, qualified installation, commissioning and clear operating guidance for residents.

Key facts and figures

Location	Strada Salubrității 42, Bucharest, Romania
Building use / function	Boutique multi-family residential building
Project type (new build, refurbishment, extension)	New build
Plot area / Floor area	Approximately 1,979 m²; public sources report 25 apartments for the certified project
Year / phase	Preliminary certification / phase reported in 2022
Climate zone	Temperate continental
Thematic focus / innovation	Healthy compact housing; mineral-wool insulation; underfloor heating; renewable electricity for common spaces; bicycle facilities
Investor / building owner	NAI Romania
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; specialist envelope and MEP suppliers
Contact person	NAI Romania / RoGBC
Further information	https://green-homes.org/property/air-residences/ https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

AIR Residence is a new build project in Strada Salubrității 42, Bucharest, Romania. It functions as boutique multi-family residential building. Its main best-practice contribution is A small residential project showing that Green Homes measures can be implemented in boutique developments, not only large estates. The documented strategy brings together mineral-wool insulation and underfloor heating designed for comfort and reduced demand, photovoltaic panels intended to offset electricity used in common areas, and secure bicycle parking and a compact four-storey urban format. RoGBC sources describe different phases with 11, 24 or 25 units; this sheet uses the current certified-project page and flags the phase difference. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Mineral-wool insulation and underfloor heating designed for comfort and reduced demand.
2. Integrated systems and operation - Photovoltaic panels intended to offset electricity used in common areas.
3. Wider value - Secure bicycle parking and a compact four-storey urban format.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.